A Project Stage-1 Report
On

DEEP LEARNING SIGNAL INTEGRITY ASSESSMENT FOR OPTICAL COMMUNICATIONS

Submitted to Kommuri Pratap Reddy Institute of Technology for the partial fulfillment of the Requirement for the Award of the Degree of

Bachelor of Technology
in
Computer Science and Engineering (AI&ML)

by

RANDHI RAM KIRAN (22RA1A6659)
MADUGU YUVARAJ (22RA1A6697)
M RAKESH (22RA1A6658)

Under the guidance of
Mrs. A. HAREESHA
M. Tech.
Assistant Professor, CSE(AI&ML)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING(AI&ML)

KOMMURI PRATAP REDDY INSTITUTE OF TECHNOLOGY

(UGC Autonomous Institution, Affiliated to JNTUH ,Hyderabad)

Ghanpur (V), Ghatkesar (M), Medchal (D)-500088

2025-2026

KOMMURI PRATAP REDDY INSTITUTE OF TECHNOLOGY

(UGC Autonomous Institution, Affiliated to JNTUH ,Hyderabad)

Ghanpur (V), Ghatkesar (M), Medchal (D)-500088

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING(AI&ML)

CERTIFICATE

This is to certify that the dissertation entitled **"DEEP LEARNING SIGNAL INTEGRITY ASSESSMENT FOR OPTICAL COMMUNICATIONS"** being submitted by **RANDHI RAM KIRAN (22RA1A6659), MADUGU YUVARAJ (22RA1A6697), M RAKESH (22RA1A6658)** in partial fulfillment for the award of Bachelor of Technology in Computer Science and Engineering (AI&ML) to the Kommuri Pratap Reddy Institute of Technology is a record of confined work carried out by them under my guidance and supervision.

Guide's signature
A.HAREESHA
M.Tech.
Assistant Professor
Dept. of CSE (AI&ML)

Signature of the Head of the Dept.
Dr. PULIME SATYANARAYANA
M.Tech.,Ph.D.
Associate Professor
Dept. of CSE (AI&ML)

External Examiner

Place: Ghanpur

Date:

KOMMURI PRATAP REDDY INSTITUTE OF TECHNOLOGY

(UGC Autonomous Institution, Affiliated to JNTUH ,Hyderabad)

Ghanpur (V), Ghatkesar (M), Medchal (D)-500088

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING(AI&ML)

DECLARATION

We, **RANDHI RAM KIRAN (22RA1A6659), MADUGU YUVARAJ (22RA1A6697), M RAKESH (22RA1A6658)** hereby declare that the project report titled **"DEEP LEARNING SIGNAL INTEGRITY ASSESSMENT FOR OPTICAL COMMUNICATIOS"** under the guidance of A. HAREESHA, Assistant Professor of Computer Science and Engineering(AI&ML), Kommuri Pratap Reddy Institute of Technology, Ghanpur, is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering (AI&ML).

This is a record of bonafide work carried out by us and the results embodied in this project have not been reproduced or copied from any source. The results embodied in this thesis have not been submitted to any other University and Institute for the award of any Degree or Diploma.

SUBMITTED BY

RANDHI RAM KIRAN	**(22RA1A6659)**
MADUGU YUVARAJ	**(22RA1A6697)**
M RAKESH	**(22RA1A6658)**

ACKNOWLEDGEMENT

The satisfaction and euphoria that accompany the successful completion of any task would be incomplete without acknowledging the individuals who made it possible.

We are extremely indebted to our internal guide, **A. HAREESHA,** Assistant Professor, for his valuable help. We would like to express our profound gratitude to her for providing us the opportunity to work under her supervision and guidance. Her guidance helped us in the successful completion of our project.

We would like to express our heartfelt gratitude to our **HOD, Dr. PULIME SATYANARAYANA, M.Tech.,Ph.D** Associate Professor , for providing the guidance and necessary infrastructure to complete this project.

We would also like to express our deep-felt gratitude to our DIRECTOR, **Dr. B. SUDHEER PREM KUMAR, M.Tech.,Ph.D** of Kommuri Pratap Reddy Institute of Technology, for his kind cooperation and valuable suggestions.

We would also like to express our deep-felt gratitude to **Dr. SRINATH KASHYAP, M.Tech.,Ph.D.,** PRINCIPAL of Kommuri Pratap Reddy Institute of Technology, for his kind cooperation and valuable suggestions.

This Acknowledgement will be incomplete without mentioning our sincere gratefulness to our honorable CHAIRMAN **Shri. KOMMURI PRATAP REDDY,** and **VICE-CHAIRMAN Shri KOMMURI PRASHANTH REDDY** of Kommuri Pratap Reddy Institute of Technology for providing the necessary infrastructure to complete this project.

We also thank the staff of the CSE(AI&ML) department who helped us morally for the successful completion of this project.

SUBMITTED BY

RANDHI RAM KIRAN	**(22RA1A6659)**
MADUGU YUVARAJ	**(22RA1A6697)**
M RAKESH	**(22RA1A6658)**

Vision of the Institute :

To emerge as a premier institute for high quality professional graduates who can contribute to economic and social developments of the Nation.

Mission of the Institute:

Mission	Statement
IM1	To have holistic approach in curriculum and pedagogy through industry interface to meet the needs of Global Competency.
IM2	To develop students with knowledge, attitude, employability skills, entrepreneurship, research potential and professionally Ethical citizens.
IM3	To contribute to advancement of Engineering & Technology that would help to satisfy the societal needs.
IM4	To preserve, promote cultural heritage, humanistic values and Spiritual values thus helping in peace and harmony in the society.

Vision of the Department:

To be a premier center of excellence with research in Artificial Intelligence and Machine Learning research through unique interdisciplinary partnerships and positive contribution to the students, community, organizations and society as a whole.

Mission of the Department:

Mission	Statement
DM1	To impart knowledge in cutting edge Artificial Intelligence technologies in par with industrial standards.
DM2	To collaborate with industry to uplift innovative research and development in Artificial Intelligence and Machine Learning and its allied fields to serve the needs of society.
DM3	To produce successful Computer Science and Engineering graduates with a specialization in AIML with personal and professional responsibilities and commitment to lifelong learning.

Program Educational Objectives (PEOs) :

PEO's	Statement
PEO1	Demonstrate technical skills, competency in AI & ML and exhibit team management capability with proper communication in a job environment.
PEO2	Support the growth of economy of a country by starting enterprise with a lifelong learning attitude.
PEO3	Carry out research in the advanced areas of Al & ML and address the basic needs of the society.
PEO4	To develop research attitude among the students in order to carry out research in cutting-edge technologies, solve real world problems and provide technical consultancy services.

PROGRAM OUTCOMES

PO1	**Engineering Knowledge:** Apply the knowledge of mathematics, science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	**Problem Analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences**.**
PO3	**Design/development of Solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	**Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	**Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	**The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	**Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental knowledge of, and need for sustainable development.

PO10	**Communication**: Communicate effectively on complex engineering activities with the engineering community and with society at large, being able to comprehend and write effective reports and design documentation, make Effective presentations, and give and receive clear instructions.
PO11	**Project management and finance**: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work,as a member and leader in a team, to manage projects and in multidisciplinary environment**.**
PO12	**Life-Long learning:** Recognize the need for, and have the preparation and able to engage in independent and life-long learning in the broadest context of technological change.
PO8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	**Individual and team network:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PROGRAM SPECIFIC OUTCOMES

PSO1	Demonstrate the knowledge of human cognition, Artificial Intelligence, Machine Learning and data engineering for designing intelligent systems.
PSO2	Apply computational knowledge and project development skills to provide innovative solutions.
PSO3	Use tools and techniques to solve problems in AI & ML.

ABSTRACT

Ensuring reliable signal transmission has become a critical requirement in modern optical and wireless communication systems, especially as data networks push toward higher bandwidth, lower latency, and greater noise resilience. Traditional signal assessment techniques depend heavily on manual thresholding or single model learning approaches, often struggling with noisy environments, imbalanced datasets, and multi-feature interactions. This creates a major gap in accurately predicting Signal Quality across diverse modulation formats, receiver parameters, and channel conditions. The proposed system addresses these limitations by introducing a data-driven, automated, and hybrid AI pipeline designed to classify signal integrity with high precision. The workflow integrates comprehensive preprocessing, label encoding, missing-value handling, and SMOTE-based class balancing to overcome skewed distributions. A suite of machine learning models including Complement Naive Bayes, Support Vector Machines, Linear Discriminant Analysis, and a custom Dual-Learner Fusion Architecture (DLFA) which trains Logistic Regression and Random Forest with Voting Classifier is trained and benchmarked using advanced metrics, confusion matrices, and ROC curves. Additionally, a fully interactive Tkinter-based GUI provides end-to-end usability, enabling dataset uploads, preprocessing, EDA visualization, model evaluation, and real-time prediction. This DLFA significantly improves decision boundaries by leveraging Random Forest's non-linear feature representations with Logistic Regression's probabilistic strengths. As a result, the system delivers enhanced robustness, improved generalization, and greater interpretability.

INDEX

S.NO	CONTENT	PAGE NO.
1.	**INTRODUCTION**	**01**
	1.1 Overview	01
	1.2 Problem Statement	01
	1.3 Motivation	02
	1.4 Significance	02
	1.5 Applications	02
2.	**LITERATURE SURVEY**	**03**
3.	**EXISTING SYSTEM**	**07**
	3.1 Traditional Systems Overview	07
	3.2 Limitations	08
4.	**PROPOSED SYSTEM**	**09**
	4.1 System Overview	09
	4.2 Architecture	09
	4.3 Tkinter-Based GUI	10
	4.4 Advantages	10
5.	**BASE PAPER**	**05**

CHAPTER 1

INTRODUCTION

1.1 Overview

Modern optical communication systems form the backbone of high-speed data transmission across global networks. As applications such as cloud computing, AI services, 5G, IoT, virtual reality, and large-scale data centres grow rapidly, the demand for reliable and high-bandwidth communication continues to increase. However, achieving error-free transmission in optical channels is challenging due to fibre nonlinearities, noise, dispersion, hardware imperfections, and varying environmental conditions.

Traditional rule-based and threshold-driven methods often fail to capture the complex and nonlinear relationships among signal parameters such as SNR, BER, power level, modulation format, and receiver characteristics. The emergence of Machine Learning (ML) and Deep Learning has introduced new opportunities for intelligent signal assessment, enabling automated, data-driven decision-making. These approaches not only enhance prediction accuracy but also offer greater adaptability to dynamic network behaviour.

1.2 Problem Statement

Optical communication systems generate large volumes of transmission data that must be continuously monitored to maintain signal quality and prevent degradation. Traditional signal assessment techniques rely heavily on fixed thresholds or single-model analysis, which cannot capture nonlinear relationships across signal features. Furthermore, class imbalance in real-world datasets causes inaccurate predictions, especially for low-quality signals.

The key problem addressed in this research is:

To develop a robust, automated, and AI-based system capable of analyzing multi-parameter optical communication data, learning hidden patterns, and accurately classifying signal quality levels under noisy and dynamic conditions.

1.3 Motivation

Several factors motivate this research:

- The exponential growth in global data traffic demands highly reliable transmission systems.
- Manual thresholding fails in high-noise environments, leading to incorrect signal quality decisions.
- Optical channels exhibit nonlinear distortion, which cannot be modeled effectively using traditional analytic methods alone.
- ML and DL-based models can adapt to dynamic channel variations and extract complex interactions between multiple signal parameters.

1.4 Significance

The proposed work is significant because it introduces:

- A fully automated end-to-end ML pipeline that reduces human dependence in signal assessment.
- A novel Dual-Learner Fusion Architecture (DLFA) that integrates Random Forest and Logistic Regression to achieve superior prediction accuracy.
- SMOTE-based class balancing to overcome skewed datasets.
- A GUI application enabling usability even for non-technical users.
- Improved network reliability through proactive quality prediction.
- Strong interpretability using confusion matrices, ROC curves, and performance metrics.

This system supports robust and generalizable signal quality evaluation across various communication domains.

1.5 Applications

The proposed Signal Integrity Assessment model is applicable in:

- Optical Fiber Networks: Monitoring SNR, BER, and dispersion-related degradations.
- Data Centers: Ensuring high-speed intra-data-center IM/DD link reliability.
- 5G/6G Backhaul Transmission: Assessing long-distance coherent optical signals.
- IoT Communication Channels: Predicting quality variations for low-power links.
- Satellite and Space Communications: Managing signal noise and fading conditions.

CHAPTER-2

2. LITERATURE SURVEY

1. Introduction

Modern high-speed electronic and optical communication systems operate at multi-gigabit data rates, where maintaining signal integrity (SI) and power integrity (PI) is essential for achieving reliable performance. As frequencies rise, channels become increasingly sensitive to crosstalk, jitter, reflections, losses, electromagnetic (EM) coupling, impedance discontinuities, and PDN noise. Traditional modeling approaches such as full-wave EM solvers, SPICE simulations, and analytical approximations offer high accuracy but at the cost of extensive computation time and limited scalability for large design spaces.

This limitation motivates the integration of machine learning (ML) and deep learning (DL) to create fast surrogate models, optimize designs, and diagnose signal-quality issues. The following review synthesizes the provided references into a structured, expanded survey covering foundations, ML advances, physics-informed methods, optimization strategies, and gaps.

2. Foundations of Signal and Power Integrity

Classical SI/PI Modeling

Textbooks by Hall & Heck [2], Bogatin [13], Pozar [4], and Swaminathan & Engin [9] form the theoretical foundation for modeling high-speed systems. These works thoroughly describe:

- Transmission line theory at microwave frequencies.
- Impedance matching, reflections, and S-parameter interpretation.
- Crosstalk mechanisms (near-end, far-end) and coupling paths.
- PDN impedance profiles, resonances, and decoupling strategies.
- Effects of dielectric materials, trace geometry, and via structures.

These principles remain crucial, as machine learning models must capture or approximate relationships governed by Maxwell's equations.

Measurement and Industry Practices

Smith & Bogatin [3] and Keysight's documentation [24] emphasize the role of equipment such as VNAs, TDR/TDT instruments, and spectrum analyzers. Accurate measurement is essential for:

- Validating ML models.
- Creating reliable datasets.
- Understanding real-world variations (temperature, process, aging).

Industry tools highlight the gap ML aims to bridge: fast what-if analysis without running heavy simulations or physical measurements repeatedly.

3. Machine Learning in Electronic Design Automation (EDA)

Survey of ML for EDA

Huang et al. [11] provide a comprehensive review, categorizing ML applications into:

- Cell-level modeling (timing, power prediction).
- Interconnect prediction (delay, capacitance, parasitic extraction).
- Layout optimization (routing, placement).
- Surrogate models for EM or reliability analysis.

The growing trend shows that ML augments traditional CAD flows by accelerating simulation and enabling large-scale optimization.

ML Surrogates for Electrical Parameters

Several works demonstrate fast surrogates that replace time-consuming numerical solvers:

- Yang et al. (CNN-Cap) [15] propose CNN models for capacitance extraction from layout patterns.

- Bansal et al. [25] predict wideband S-parameters using deep networks.

- Ren et al. [20] (ParaGraph) use graph neural networks to model parasitics—well suited because PCB/IC layouts inherently have graph structures.

These approaches typically offer *1,000x faster inference* with minimal accuracy loss.

Application-Specific ML Models in SI/PI

- Crosstalk estimation using ML (Shi et al. [14]).
- Jitter decomposition using CNNs (Ren et al. [17]).
- Partial discharge detection using pattern-recognition (Long et al. [22]).
- Component-level modeling and optimization for PDNs (Jiang et al. [10]).

This demonstrates ML's versatility across waveform classification, parameter regression, and structural optimization.

4. Physics-Informed and Hybrid ML Approaches

Physics-Informed Neural Networks (PINNs)

PINNs integrate physical laws into neural network training. Unlike black-box models, they enforce PDE constraints such as Maxwell's equations.

Qi & Sarris [8] introduce unconditionally stable PINNs for wave equations, showing:

- Better generalization under limited training data.
- Greater stability for high-frequency problems.
- Direct incorporation of boundary/initial conditions.

Khan & Lowther [19] apply PINNs in electromagnetics, highlighting their ability to model field distributions where data is scarce.

Hybrid Physics + ML Optimization

Hybrid strategies merge fast ML predictions with physics-based heuristics or solvers:

- Jiang et al. [10] integrate physics-assisted genetic algorithms for capacitor optimization.
- Song & Sui [26] discuss ML-assisted frameworks for SerDes optimization.
- Vega & Chevrier [23] outline co-simulation-based optimization combining various solvers.

Hybridization is especially important for ensuring reliability when ML models replace parts of traditional design flows.

5. Applications of ML to SI/PI Challenges

Crosstalk Prediction and Mitigation

High-density PCBs require fast prediction of coupling. ML methods:

- Learn spatial features from trace arrangements.
- Offer near-instantaneous estimation versus solver runtimes.
- Enable routing exploration during early design phases.

Jitter Breakdown and Signal Quality Classification

DL-based jitter decomposition (Ren et al. [17]) helps engineers understand:

- Deterministic jitter components (ISI, crosstalk-induced).
- Random jitter sources.
- Potential SI violations.

Parametric Electrical Modeling

CNNs and GNNs generate models for:

- Return paths.
- Interconnect impedance.
- Capacitance/resistance matrices.
- S-parameter frequency responses.

These outputs are essential for SerDes, memory interfaces, high-speed buses, and optical receivers.

The intersection of signal/power integrity engineering and machine learning is rapidly evolving. Foundational EM and SI/PI theory remain essential, but deep learning, graph neural networks, and physics-informed models significantly accelerate prediction and optimization. The reviewed works demonstrate substantial progress in parasitic prediction, crosstalk and jitter modeling, PDN optimization, and wave-equation-based surrogate modeling.

CHAPTER - 3

3. EXISTING SYSTEM

3.1 Traditional System Overview :

Traditional signal quality analysis in optical communication systems relies on conventional, threshold-based, and rule-driven techniques. These systems typically use fixed thresholds for SNR, BER, OSNR, and power levels to assess whether a signal is "Good," "Average," or "Poor.

Characteristics of Existing Systems

Existing systems follow these steps:

- Manual observation of communication parameters
- Threshold-based decision-making
- Rule-based classification
- Statistical estimation methods
- Fixed-parameter boundaries
- Limited use of automation

This process depends heavily on human expertise and is prone to frequent errors—especially in real-time monitoring environments.

Limitations of the Existing System

1. **Inability to handle nonlinear interactions**
   Traditional systems fail to capture complex relationships between parameters such as Tx, Rx, modulation format, and noise.
2. **Poor performance in noisy environments**
   Thresholds become inaccurate when data is distorted by noise, interference, or channel fluctuations.
3. **High dependency on expert knowledge**
   Skilled engineers must participate in diagnosis, making the system slow and inefficient.

4. **No support for imbalanced datasets**
   Minority classes—often the most critical—are ignored by basic threshold-based models.

5. **Lack of scalability**
   As networks expand, manual monitoring becomes impossible.

6. **Slow decision-making**
   Thresholding cannot adapt to dynamic real-time variations in high-speed systems.

The limitations clearly show that traditional systems are insufficient for modern, high-speed communication environments.

CHAPTER - 4

4. PROPOSED SYSTEM

The proposed system introduces a fully automated, AI-powered pipeline for highly accurate Signal Quality classification in optical communication systems. It integrates a comprehensive preprocessing workflow, multiple ML models, and a uniquely designed fusion architecture for improved accuracy and robustness.

4.1 System Overview

The system consists of:

- Data ingestion and preprocessing
- Missing-value handling
- Label encoding
- SMOTE-based class balancing
- Training of multiple ML models
- Implementation of the **Dual-Learner Fusion Architecture (DLFA)**
- Evaluation using ROC, confusion matrix, and accuracy metrics
- GUI-based user interaction
- Real-time prediction module

This provides an end-to-end solution that is automated, scalable, and highly precise.

4.2 Dual-Learner Fusion Architecture (DLFA)

DLFA is the core innovation of the proposed system. It integrates:

Logistic Regression (LR)

- Strong in probabilistic classification
- Creates linear decision boundaries
- Stable and interpretable

Random Forest (RF)

- Handles nonlinear relationships
- Learns multi-dimensional patterns

Voting Classifier

- Combines predictions from LR and RF
- Produces final fused output
- Improves accuracy and generalization

This hybrid approach merges the strengths of linear and nonlinear learners, resulting in improved performance.

4.3 Tkinter-Based GUI

A user-friendly interface enables:

- Dataset upload
- Preprocessing
- EDA visualization
- Model training
- Accuracy and performance display
- ROC and confusion matrix plotting
- Real-time prediction on new datasets

This allows non-technical users to perform full ML operations easily.

4.4 Advantages of the Proposed System

- Fully automated ML pipeline
- Superior accuracy due to model fusion
- Handles class imbalance using SMOTE
- Robust against noise and nonlinearities
- GUI enables easy real-time usage
- Supports end-to-end deployment

Artificial Intelligence Applications to Enhance Signal and Power Integrity in High-Speed Systems

Kunal Kumar
Electrical Engineering,National Cheng Kung University Tainan Taiwan

Signal integrity (SI) and power integrity (PI) have become more challenging to maintain as a result of the continuous expansion of high-speed electronic systems in fields like as 5G, high-performance computing, and automotive electronics. Although traditional electromagnetic simulations and optimization techniques can be accurate, they are slow and expensive. Recent developments in artificial intelligence (AI), particularly in the areas of machine learning and deep learning, present new opportunities to accelerate SI/PI analysis, prediction, and design. Crosstalk, return loss, impedance, power distribution network impedance, and voltage fluctuations may all be quickly estimated thanks to AI models' comprehension of the intricate relationships between design elements and system behavior.Decoupling capacitor placement is also made smarter by optimization techniques based on evolutionary computing and reinforcement learning.Create adaptive PDNs and combine SI and PI optimization.AI is a viable approach for future high-speed systems since it enhances dependability and cuts down on design times when added to electronic design automation (EDA) processes.

keywords:
ArtificialIntelligence,SignalIntegrityPowerIntegrity,InsertionandReturnLoss,PDNImpedance,Electronic Design Automation (EDA)

I. Introduction

In high-speed electronic systems, the constant demand for increased data rates, reduced latency, and improved energy efficiency has increased the significance of Signal Integrity (SI) and Power Integrity (PI) studies. Applications in cloud data centers, 5G communication, high-performance computing, and automotive electronics require power delivery networks (PDNs), printed circuit boards (PCBs), and interconnects to operate at multi-gigahertz rates with consistent and dependable performance[1],[2].Parasitic phenomena such crosstalk, reflections, ground bounce, voltage droop, and simultaneous switching noise, however, become significant variables that impair system performance at high rates.

Conventional SI/PI analysis relies on heuristic optimization methods, SPICEbased circuit modeling, and full-wave electromagnetic (EM) simulations. These techniques provide insightful information on the behavior of the system. But they can require a lot of resources, take a long time, and require a lot of experience[3].

Simulation time and design costs increase dramatically with system complexity, particularlywhenmulti-layerPCBsand2.5D/3Dintegratedcircuitsareincluded.The design cycle is delayed as a result [4]. Artificial Intelligence (AI), particularly machinelearning(ML)anddeeplearning(DL),hasemergedasapotentinstrumentto addresstheseissues. ComplexandnonlinearcorrelationsbetweenSI/PIperformance

measures and design parameters can be learned by AI systems. Consequently, fewer full-scale simulations are required [5]-[6]. For example, surrogate ML models are able to rapidly and accurately predict ocular diagram features, impedance mismatches, and S-parameters [7]. Rapid characterisation has also been made possible by the use of deep neural networks to reconstruct ocular diagrams and signal waveforms from sparse measurement data [8].

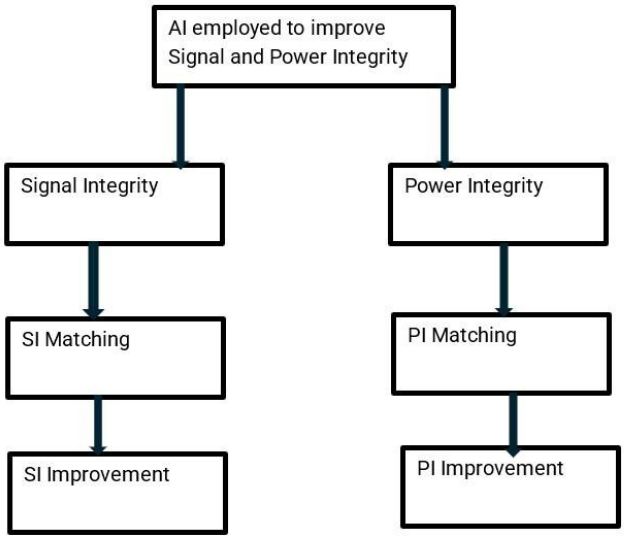

Fig. 1 Block diagram of AI based SI PI .

AI has demonstrated potential in the field of power integrity by enhancing the placement of decoupling capacitors, modifying PDN impedance, and forecasting transient voltage droop [9]. Additionally, researchers have looked into genetic algorithms and reinforcement learning for co-optimizing power integrity and signal integrity. This aids designers in balancing the trade-offs between power delivery efficiency and signal quality [10]. Adaptive optimization, anomaly detection, and automated design rule verification are other benefits of integrating AI into Electronic Design Automation processes. The time it takes to introduce sophisticated electrical devices to the market is shortened by these developments [11]. In order to increase the precision, dependability, and credibility of AI-assisted SI/PI analysis, hybrid techniques that blend physics-based modeling with AI-driven prediction are showing promise [12].

The applications of AI to enhancing signal and power integrity in high-speed systems are reviewed in detail in this work. Impedance modeling, jitter analysis, and crosstalk prediction are examples of AI-driven techniques for SI analysis that are covered in Section 2. PDN optimization and voltage noise prediction are two examples of AI applications in PI that are examined in Section 3.The AI-assisted SI/PI co-design techniques are highlighted in Section 4. Case studies and new trends are covered in Section 5. Section 6 concludes by outlining present issues and potential avenues for further study.

II. Signal Integrity (SI) with Artificial Intelligence AI

The signal integrity (SI) of high-speed systems has a significant impact on their performance. The ability of sent signals to move without distortion, delays, or loss is determined by this factor. SI issues such as insertion loss, jitter, crosstalk, reflections, and eye closure become more significant when data rates increase to multi-gigabit per second levels [2]-[13]. Transmission line models, circuit simulations, and full-wave electromagnetic (EM) solvers are the mainstays of traditional SI analysis. Though they are expensive to compute and frequently impracticable in the early phases of design, these methods yield accurate forecasts[7]. Machine learning (ML) and deep learning (DL), two recent developments in artificial intelligence (AI), offer a possible substitute. While maintaining high accuracy, they can expedite SI analysis.

A. Declaration of Crosstalk

A major problem with high-speed PCB and package interconnects is crosstalk, both near-end (NEXT) and far-end (FEXT). The geometry of the connection, the material's characteristics, and the switching patterns all affect its level. Large design spaces make traditional EM-based analysis unworkable since it requires re-simulation for each design modification [4].

Using geometry and material data, machine learning algorithms are able to anticipate crosstalk with high speed. To anticipate NEXT and FEXT across frequency ranges [14], for example, simulation data can be used to train support vector regression (SVR) and random forest algorithms. Additionally, it has been demonstrated that deep neural networks (DNNs) can comprehend intricate nonlinear correlations between crosstalk levels and layout variables like trace width, spacing, and layer stack-up. By incorporating signal integrity awareness into PCB layout

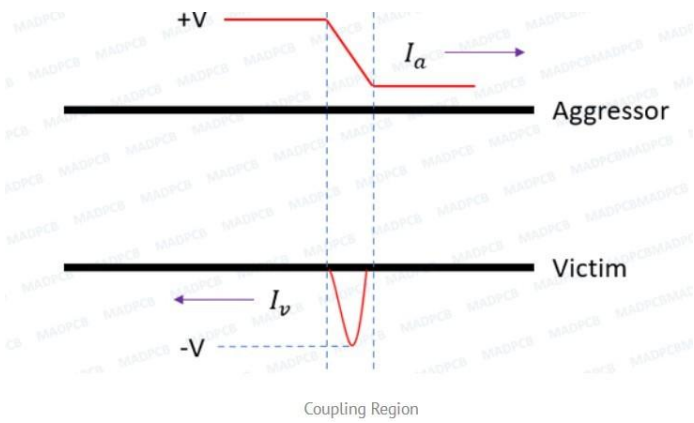

Fig. 2 Crosstalk in signal transmission .

design, automated routing techniques can significantly reduce the number of design iterations.

B. Sustaining Reflection and Impedance

Signal quality is decreased and reflections are produced when drivers, transmission lines, and loads have different impedances. Typically, modeling requires the extraction of frequency-domain S-parameters using electromagnetic simulations.

Surrogate models powered by AI provide a more effective option. For instance, convolutional neural networks (CNNs) assist in reducing repeated solver calls by directly connecting layout parameters to impedance profiles [15]. Similarly, with limited training data, Gaussian Process Regression (GPR) models forecast reflection coefficients over a broad frequency range [16].

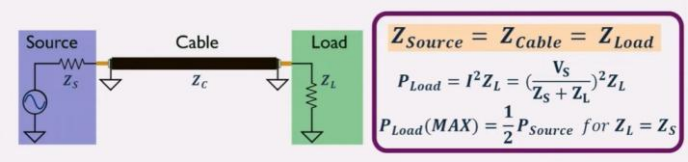

Fig. 3 Impedance matching in transmission lines .

Automaticdesignofimpedancematchingnetworksismadepossiblebycombining AI-powered quick impedance predictors with optimization techniques such as particle swarm optimization (PSO) and genetic algorithms (GA) . Quick parameter space exploration is made possible by this combination, which lowers return loss and boosts transmission efficiency.

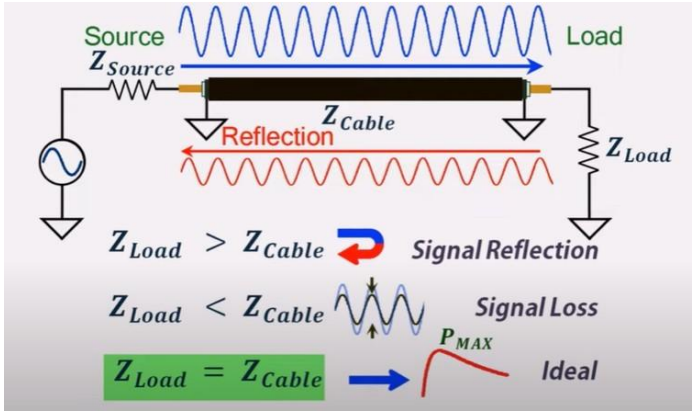

Fig. 4 Reflection in signal transmission .

C. Jitter and Eye Diagrams Analysis

Inter-symbol interference (ISI), crosstalk, and power supply noise are some of the causes of jitter. In high-speed systems, it is a crucial signal integrity parameter that influences timing margins. Conventional techniques for jitter reduction depend on in-depth time-domain simulations or measurements [9].

AI has demonstrated significant potential in the rapid prediction and classification of jitter. Long Short-Term Memory (LSTM) models and recurrent neural networks (RNNs) both examine time-series data of signal waveforms. They are able to recognize deterministic jitter (DJ), bounded uncorrelated jitter (BUJ), and random jitter (RJ) components automatically [17].

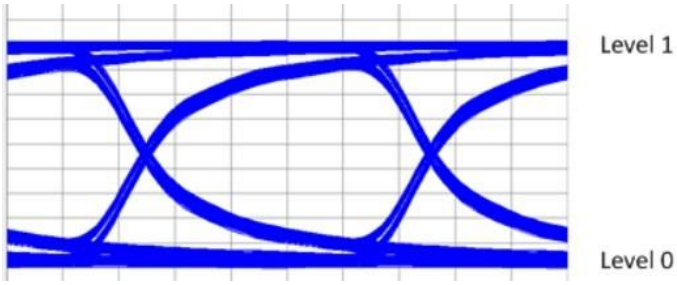

Fig. 5 Impedance matching in transmission lines .

Time-consumingbit-error-rate(BER)simulationsaretypicallyneededtogenerate eye diagrams. AI can expedite this process, though. Accurate estimation of opening height and width and reconstruction of entire eye diagrams are possible using deep learning models trained on sparse signal data. In hardware testing environments, this enables real-time monitoring and significantly reduces the calculation time required to certify high-speed communications.

D. S-Parameter Modeling and Learning

In frequency domain analysis, S-parameters are frequently employed to characterize high-speed interconnects. Typically, costly electromagnetic simulations or vector network analyzer (VNA) measurements are needed to create S-parameters. Based on geometry and material inputs, AI-based surrogate models, such Gaussian process regression (GPR) and deep neural networks (DNNs), can efficiently estimate S-parameter frequency responses [18]. These models allow designers to evaluate the consequences of changing parameters without having to rerun solvers since they reduce computation time and facilitate parametric sensitivity analysis.

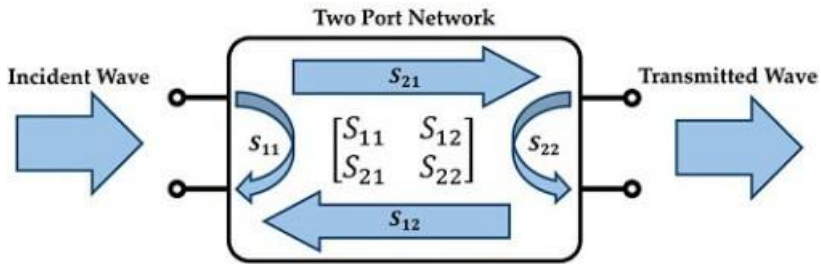

Fig. 6 S parameter matrix.

Improved physical consistency and generalization have been demonstrated by hybrid techniques that integrate AI models with physics-based knowledge[19]. For example, during training, Maxwell's equations are used by physics-informed neural networks (PINNs). As a result, the surrogate models are more in line with electromagnetic theory.

E. Design Rule Checking (DRC) for SI

A crucial stage in PCB design is Design Rule Checking (DRC).It assists in making sure that designs adhere to signal integrity (SI) requirements. Set rules are used in traditional DRC techniques, which may not be able to capture intricate interactions at high data rates. DRC technologies driven by AI are able to learn from vast collections of successful and unsuccessful designs. They identify SI infractions automatically. For example, PCB traces are represented as graphs using graph neural networks (GNNs) [20]. This makes it possible

to detect SI breaches that take the topology of the design into account effectively. Early in the design phase, these AI techniques can spot possible signal degradation problems. This lessens the requirement for costly rework.

AI has shown considerable potential in SI applications like as automating DRC, assessing jitter, modeling impedance, predicting crosstalk, and reconstructing eye diagrams. Artificial Intelligence reduces computational costs, expedites design closure, and enhances system reliability by supplementing or replacing conventional solver-based approaches. Making decisions in real time during routing and verification is also made easier by the integration of AI into EDA procedures. Problems like interpretability, generalization across different technologies, and dataset availability, however, remain active study fields.

III. Power Integrity (PI) with AI

The goal of Power Integrity (PI) is to provide steady power to the entire system, whereas Signal Integrity (SI) guarantees clear signal delivery. Because contemporary high-speed systems operate at lower supply voltages and higher current densities, they are more susceptible to simultaneous switching noise (SSN), ground bounce, voltage drops, and power supply noise [3]. A wide frequency range of noise reduction depends on the Power Distribution Network (PDN), which consists of PCB planes, decoupling capacitors, package interconnects, and voltage regulators [9].

Frequency-domain impedance extraction or transient simulations are required for traditional PI analysis. For large designs, these approaches require a lot of resources [4]. AI provides a data-driven approach that enables flexible PDN design, transient noise forecasting, faster impedance estimate, and more intelligent decoupling capacitor management. These techniques expedite the design process and reduce the requirement for thorough EM simulations.

A. Modeling PDN Impedance

Throughout the operational frequency range, maintaining the PDN impedance below the target impedance is a primary objective of PI analysis [7]. Solvers that are based on SPICE or EM are used in traditional extraction.
Artificial intelligence-based surrogate models offer effective PDN impedance approximations. Decoupling capacitor placement, material characteristics, and geometric parameters have all been linked to frequency-domain impedance profiles using deep neural networks (DNNs) and convolutional neural networks (CNNs) [16]. Predicting impedance under parameter fluctuations is made easier with the help of Gaussian Process Regression (GPR) models, which provide uncertainty quantification .

PINNs, or Physics-Informed Neural Networks, have been used in recent research. In this method, the training procedure incorporates circuit limitations and Maxwell's

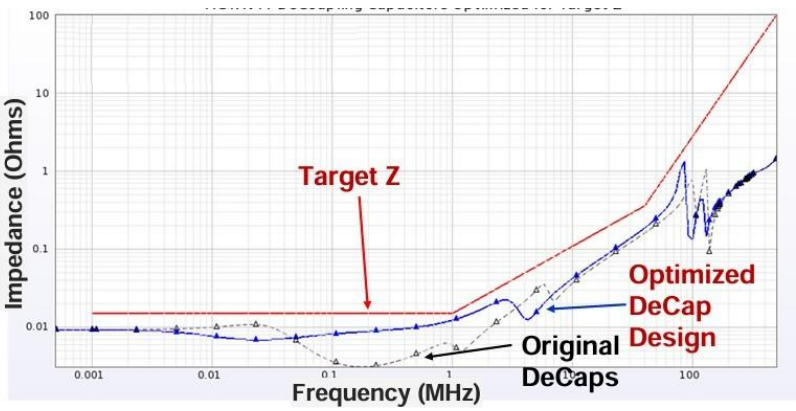

Fig. 7 Target impedance optimization with decoupling capacitor.

equations, which helps maintain the physical consistency of the AI models[19]. Large datasets are not as necessary with this approach, which also enhances generalization across various PDN topologies.

B. Optimization of Decoupling Capacitors

Decaps, also known as decoupling capacitors, are essential for lowering noise in the power distribution network (PDN) and stabilizing supply voltages. Finding the ideal capacitor values, amounts, and locations to reduce impedance peaks and satisfy the desired impedance requirements is difficult [2].

Conventional approaches use several circuit simulations in conjunction with optimization techniques or design criteria. This procedure is considerably accelerated by AI-based techniques. Fast AI-based impedance predictors combined with particle swarm optimization and genetic algorithms allow for effective exploration of the vast solution space [3]. Additionally, reinforcement learning agents have the ability to dynamically select capacitor configurations. Comparing this to conventional approaches, improved trade-offs between cost and performance are achieved .

C. Forecast for Ground Bounce and Voltage Droop

Large current spikes are caused when switching numerous drivers simultaneously at high data speeds. Ground bounce and voltage droop are the results of this. In addition to harming timing margins, these problems may lead to system failures [9]. Long Short-Term Memory (LSTM) networks and Recurrent Neural Networks

(RNNs) are two AI time-series models that can forecast supply droops during switching patterns [21]. Design modifications and real-time solutions are aided by these models.

Supervised learning techniques can also determine the primary sources of

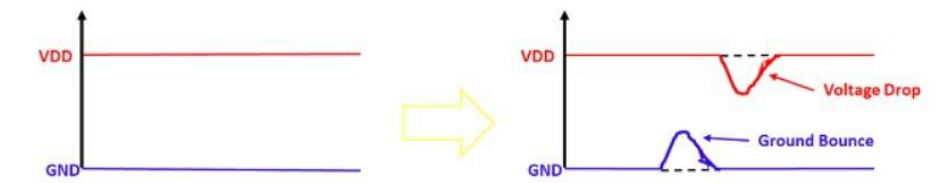

Fig. 8 Target impedance optimization with decoupling capacitor.

noise and the degree of droop. Early detection of power distribution network vulnerabilities is facilitated by this[22]. AI models eliminate the requirement for recurrent transient simulations by utilizing historical simulation or measurement data.

D. Integration of SI and PI

Because SI and PI are closely related, changes made to one's design frequently have an impact on the other. Capacitor placement for PI, for instance, may change return current routes, creating additional SI problems[13]. Co-optimization is difficult to do with conventional sequential procedures.

AI makes multi-objective optimization easier by taking SI and PI performance metrics into account at the same time. Combining evolutionary algorithms with surrogate models allows them to investigate design spaces for the best possible trade-offs between PDN impedance, jitter, and insertion loss [23]. Additionally, shared AI models can jointly forecast SI and PI measures through multi-task learning (MTL) techniques, offering reliable system-level optimization .

PDN impedance modeling, decoupling capacitor optimization, voltage droop prediction, and adaptive control techniques are a few of the AI applications in PI. AI increases the effectiveness and precision of PI analysis through the use of multiobjective optimization, reinforcement learning, and surrogate modeling. These techniques lower processing requirements and enable clever, flexible PDN designs that improve robustness in high-speed systems of the future. Nonetheless, issues such guaranteeing physical interpretability, diversity of datasets, and generalization across technologies remain significant research topics.

IV. AI-Powered Co-Design Methods

High-speed systems are intimately related to both Signal Integrity (SI) and Power Integrity (PI). Current return routes can be impacted and crosstalk can be increased by modifications to the power distribution network (PDN), such as the location of decoupling capacitors. Likewise, PDN impedance may change and supply noise may worsen if routing or stack-up changes are made for SI optimization [13]. Historically, independent design approaches have been used to examine SI and PI. When problems occur, SI and PI teams work independently to try to resolve them. Long verification cycles, less-than-ideal trade-offs, and design inefficiencies are common outcomes of this approach [7].

A useful tool for developing co-design techniques that jointly maximize SI and PI constraints is artificial intelligence (AI).AI-powered techniques assist designers in resolving conflicting needs and achieving high performance in a variety of domains through the use of multi-objective optimization, surrogate modeling, and adaptive learning.

A. Frameworks for Multi-Objective Optimization

Conventionaloptimizationforpowerintegrity(PI)orsignalintegrity(SI)typically focuses on just one objective. It frequently seeks to lower power distribution network (PDN) impedance or minimize insertion loss. High-speed systems, on the other hand, require multi-objective optimization (MOO), which simultaneously takes into account jitter, eye height, crosstalk, power noise, and area/cost constraints [21].

Multi-objective SI/PI optimization has successfully employed evolutionary algorithms (EA), such as particle swarm optimization (PSO) and genetic algorithms (GA). These algorithms can effectively explore vast design areas without requiring full simulations for each iteration when combined with AI-based surrogate models. AI-accelerated MOO, for example, can reduce PDN impedance and return loss, guaranteeing that the SI and PI objectives are met simultaneously [1]. Recent studies have also examined co-design based on reinforcement learning (RL). With this method, agents learn how to balance SI and PI measurements while dynamically modifying routing, decap placement, and stack-up parameters. In complicated system-on-package (SoP) and 3D integrated circuit (IC) environments,

where dependencies are more noticeable, this adaptable optimization technique is

particularly advantageous.

B. AI for Optimizing Routing and Stack-Up

Power integrity and signal integrity are greatly impacted by PCB and packaging stack-up design. Power distribution network performance, signal loss, and impedance control are all directly impacted by variables including plane distribution, copper roughness, and dielectric thickness [2]. Optimizing stack-up has historically been a laborious, manual procedure.

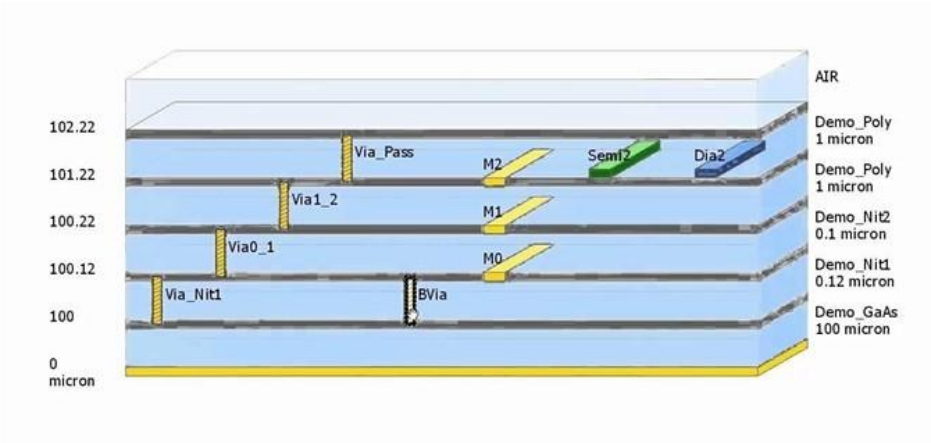

Fig. 9 PCB layer stackup with keysight ads.

AI-based optimization methods help simplify stack-up design by figuring out how material and geometry factors [24] relate to power and signal integrity measures [25]. Reinforcement learning-based routing techniques improve this process even more by real-time modifying return pathways, trace widths, and spacing to lessen ground bounce and crosstalk. One particularly intriguing technique for co-design is the use of graph neural networks. Power distribution networks and PCB interconnects are able to capture topological information since they can be represented as graph structures.

C. Examples of AI-Powered Co-Design Cases

Numerous recent research demonstrate AI's potential for SI/PI co-design:

High-Speed SerDes Channels: PDN impedance and channel loss were optimized using DNN-based surrogate models. When compared to conventional procedures, our method reduced simulation time by 70% [26].

3D IC Packages: An automatically balanced through-silicon via (TSV) placement reinforcement learning framework. As a result, PDN impedance peaks and coupling noise were reduced [27].

PCB Decap Placement: CNN-based impedance predictors combined with evolutionary algorithms discovered decap designs that reduced PDN impedance

without compromising return channel integrity .

These studies demonstrate how AI-enabled co-design frameworks can enhance SI/PI trade-offs and drastically reduce design cycles.

The main obstacles to AI-enhanced SI/PI co-design remain despite advancements: Data Scarcity Large datasets from EM simulations or measurements are necessary for model training. It is expensive to create these datasets.

Generalization: AI models that were trained on FR4 PCBs or another technology might not perform well on advanced packaging or other technologies.

Interpretability: Designers must comprehend the rationale behind AI's recommended trade-offs. But the majority of models are black boxes.

Integration with EDA Tools: There has been little progress in integrating AI models into commercial design environments. Transfer learning for crosstechnology generalization, explainable AI (XAI) to facilitate more adoption in mission-critical applications, and hybrid AI-physics techniques are anticipated to be the main areas of future study [17].

V. New Developments

AI's application in Signal Integrity (SI) and Power Integrity (PI) design has progressed from concepts to practical uses. Clear gains in design time, accuracy, and dependability are demonstrated in a number of case studies. At the same time, emerging trends in packaging technologies and Electronic Design Automation (EDA) suggest that AI-driven co-design will play a bigger role in electronic systems of the future.

A. PCB Decap Placement Optimization

Placing decoupling capacitors (decaps) is one of the most difficult power integrity issues. Poor impedance profiles or over-design are common outcomes of traditional rule-based approaches [2]. Convolutional neural networks (CNNs) and evolutionary algorithms combined in AI-enabled frameworks have lower power delivery network (PDN) impedance and require fewer capacitors[22]. In order to determine effective locations that preserve return path continuity, the AI system learns the impedance characteristics of various configurations. This lessens ground bounce-induced loss of signal integrity (SI).

B. High-Speed Channels for SerDes

Channel loss, jitter, and crosstalk must be strictly controlled for SerDes (serializerdeserializer) channels at data speeds more than 56 Gb/s. Using S-parameter datasets as training data, a case study demonstrated that deep neural networks (DNNs) can reliably predict insertion loss and return loss across various stack-ups [10].

The surrogate model reduced simulation cycles by 70% as compared to conventional full-wave solvers, enabling rapid "what-if" scenarios [4]. AI can be a helpful

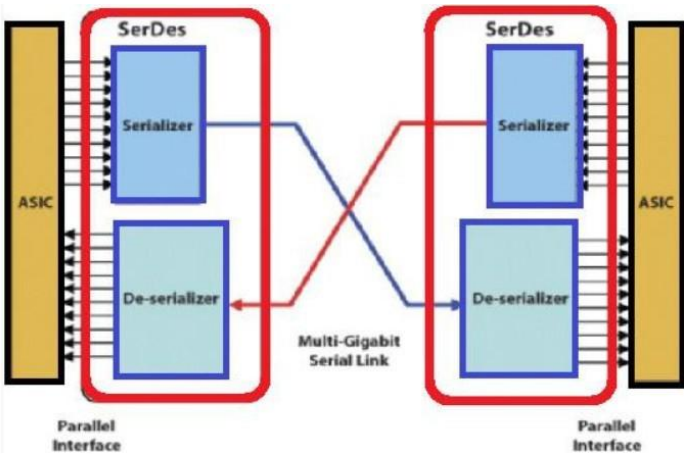

Fig. 10 High-speed SerDes interfaces.

tool prior to costly electromagnetic (EM) studies, as this method shows.

C. Advanced Packaging and 3D IC

As 3D ICs and heterogeneous integration have become more common, throughsiliconvias(TSVs), micro-bumps, andredistributionlayers(RDLs)havecomplicated the relationship between signal integrity (SI) and power integrity (PI) [20]. Researchers have improved TSV placement by balancing power delivery network (PDN)

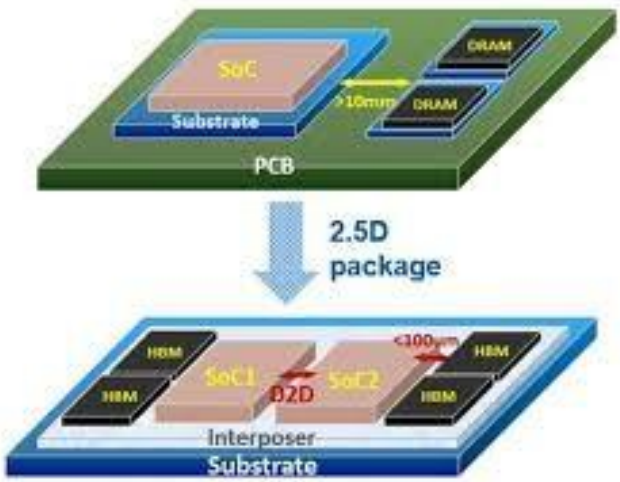

Fig. 11 3D-IC Chip design .

impedance, which is a PI concern, and coupling noise, which is a SI concern, using reinforcement learning (RL) techniques [27]. With a decreased noise coupling and PDN stability, the RL-based framework performed better than the heuristic approach, proving the usefulness of AI in optimizing several aspects of advanced packaging.

D. EDA Workflows in Integration

To provide design assistants who make real-time recommendations for routing patterns, stack-up alternatives, or decap tactics, major EDA suppliers are investigating AI integration

[17]. Closed-loop workflows are becoming more popular. AI models are immediately integrated into layout tools in these workflows, continuously monitoring SI/PI parameters and providing remedial input while the design is being entered. This signifies a change from post-layout verification to co-optimization in real time.

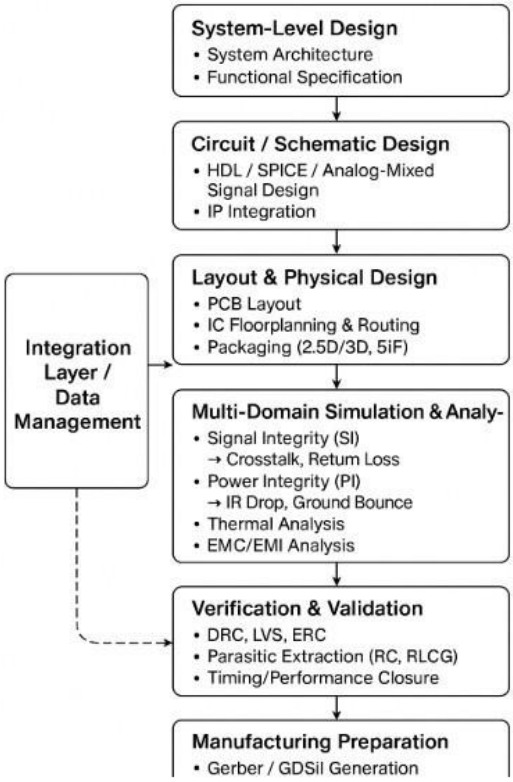

Fig. 12 EDA Integration flow.

The importance of SI/PI co-design will increase when data rates in 6G, data center, and automotive applications approach 112 Gb/s and higher. AI techniques need to advance not only in prediction but also in process comprehension, generalization, and automation. Future studies are probably going to focus on: multi-area cooptimization, including mechanical, thermal, PI, and SI stress.Companies can train models together without exchanging sensitive data thanks to federated learning techniques.frameworks for autonomous EDA in which AI-powered agents assess design alternatives with minimal assistance from humans. The transition from reactive checking to proactive, intelligent co-design is ultimately signified by AI-driven SI/PI design. Stronger and more affordable designs for the upcoming generation of high-speed systems will result from this.

VI. Conclusion

This study has looked at the rapidly expanding application of artificial intelligence (AI) to address problems with power integrity (PI) and signal integrity (SI) in highspeed electronic systems. Design guidelines based on experience and expensive full-wave simulations are key components of traditional SI/PI analysis. Frequently, these techniques are unable to keep up with the growing complexity of dense PCBs, multi-gigabit SerDes channels, and sophisticated package designs. AI methods have demonstrated significant potential in accelerating analysis, enhancing design parameters, and raising predicting accuracy.

Examples of these methods include deep learning, reinforcement learning, graph-based learning, and hybrid physicsinformed models. There are still a number of unexplored study areas. First, in order to integrate SI, PI, thermal, and mechanical effects inside unified optimization frameworks, AI models must be expanded to multi-physics co-design. Second, leveraging varied industrial datasets while adhering to proprietary limits will require federated and privacy-preserving learning. Third, enhancing explainability would ensure that AI forecasts are not just accurate but also understandable, boosting designer trust. Finally, a significant move toward intelligent, automated electronic system design is shown by the development of autonomous EDA agents, which can investigate design options with little assistance from humans. In conclusion, engineers may approach SI/PI problems in a new way thanks to AI. The future lies in AI-powered design intelligence rather than solely depending on simulationdriven verification. In order to satisfy the increasing demands of next-generation high-speed systems, this method offers both speed and precision.

References

[1] Ya, V., Chen, Y. C., Chou, Y. H., Choo, K. H., Liu, J. C., Mameda, N., and Li, C. W., "Cryolite (Na 3 AlF 6) crystallization for fluoride recovery using an electrolytic process equipped with a sacrificial aluminum anode," *Journal of Hazardous Materials*, Vol. 368, 2019, pp. 90–96. https://doi.org/10.1016/j.jhazmat.2019.01.033, URL https://doi.org/10.1016/j.jhazmat.2019.01.033.

[2] Hall, S. H., and Heck, H. L., *Advanced signal integrity for high-speed digital designs*, John Wiley & Sons, 2011.

[3] Smith, L., and Bogatin, E., "Principles of power integrity for PDN design," *2018 IEEE Symposium on Electromagnetic Compatibility, Signal Integrity and Power Integrity (EMC, SI & PI)*, IEEE, 2018, pp. 1–74.

[4] Pozar, D. M., "Microwave engineering," *Fourth Editions, University of Massachusetts at Amherst, John Wiley & Sons, Inc*, 2012, pp. 26–30.

[5] LeCun, Y., Bengio, Y., and Hinton, G., "Deep learning," *nature*, Vol. 521, No. 7553, 2015, pp. 436–444.

[6] Keutzer, K., Li, P., Shang, L., and Zhou, H., "ACM transactions on design automation of electronic systems (TODAES) special section call for papers: parallel CAD: algorithm design and programming," *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, Vol. 15, No. 1, 2009, pp. 1–2.

[7] Johnson, H., et al., *High speed digital design: a handbook of black magic*, Pearson Education India, 1993.

[8] Qi, S., and Sarris, C. D., "Hybrid physics-informed neural network for the wave equation with unconditionally stable time-stepping," *IEEE Antennas and Wireless Propagation Letters*, Vol. 23, No. 4, 2024, pp. 1356–1360.

[9] Swaminathan, M., and Engin, E., *Power integrity modeling and design for semiconductors and systems*, Pearson Education, 2007.

[10] Jiang, L., Zhang, L., Tan, S., Li, D., Hwang, C., Fan, J., and Li, E.-P., "A novel physics-assisted genetic algorithm for decoupling capacitor optimization," *IEEE Transactions on Microwave Theory and Techniques*, Vol. 72, No. 8, 2024, pp. 4498–4507.

[11] Huang, G., Hu, J., He, Y., Liu, J., Ma, M., Shen, Z., Wu, J., Xu, Y., Zhang, H., Zhong, K., et al., "Machine learning for electronic design automation: A survey," *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, Vol. 26, No. 5, 2021, pp. 1–46.

[12] Samek, W., Montavon, G., Vedaldi, A., Hansen, L. K., and Müller, K.-R., *Explainable AI: interpreting, explaining and visualizing deep learning*, Vol. 11700, Springer Nature, 2019.

[13] Bogatin, E., *Signal and power integrity–simplified*, Pearson Education, 2010.

[14] Shi, D., Sun, N., Liu, Y., Lian, C., Chen, X., Zhou, X., Liu, X., Liu, Q., and Zhou, J., "Fast calculation method for crosstalk of high-density printed circuit board by using machine learning," *IEEE Transactions on Electromagnetic Compatibility*, Vol. 66, No. 3, 2024, pp. 971–982.

[15] Yang, D., Li, H., Yu, W., Guo, Y., and Liang, W., "CNN-Cap: Effective convolutional neural network-based capacitance models for interconnect capacitance extraction," *ACM Transactions on Design Automation of Electronic Systems*, Vol. 28, No. 4, 2023, pp. 1–22.

[16] Lei, Y., Guo, M., Cai, H., Hu, D., and Zhao, D., "Prediction of length-of-day using Gaussian process regression," *The Journal of Navigation*, Vol. 68, No. 3, 2015, pp. 563–575.

[17] Ren, N., Fu, Z., Zhou, D., Liu, H., Wu, Z., and Tian, S., "Jitter decomposition by convolutional neural networks," *IEEE Transactions on Electromagnetic Compatibility*, Vol. 63, No. 5, 2021, pp. 1550–1561.

[18] Song, Y., Shen, C., and Liu, X., "A surrogate model for shallow water equations solvers with deep learning," *Journal of Hydraulic Engineering*, Vol. 149, No. 11, 2023, p. 04023045.

[19] Khan, A., and Lowther, D. A., "Physics informed neural networks for electromagnetic analysis," *IEEE Transactions on Magnetics*, Vol. 58, No. 9, 2022, pp. 1–4.

[20] Ren, H., Kokai, G. F., Turner, W. J., and Ku, T.-S., "ParaGraph: Layout parasitics and device parameter prediction using graph neural networks," *2020 57th ACM/IEEE Design Automation Conference (DAC)*, IEEE, 2020, pp. 1–6.

[21] Lee, J.-W., Park, K.-J., Cho, J., Kim, J.-Y., and Son, S.-Y., "Novel monitoring system for low-voltage DC distribution network using deep-learning-based disaggregation," *IEEE Access*, Vol. 8, 2020, pp. 185266–185275.

[22] Long, J., Xie, L., Wang, X., Zhang, J., Lu, B., Wei, C., Dai, D., Zhu, G., and Tian, M., "A comprehensive review of signal processing and machine learning technologies for UHF PD detection and diagnosis (II): Pattern recognition approaches," *IEEE Access*, Vol. 12, 2024, pp. 29850–29890.

[23] Vega, D. A., and Chevrier, V., "Co-simulation-based optimization: state of the art," 2024.

[24] Keysight, X., and Anayzers, S. S., "Keysight technologies," *Santa Rosa, CA, USA*, 2014.

[25] Bansal, V., Feng, L., de la Rubia, V., and Benner, P., "Parametric S-Parameter Prediction Using Deep Learning," *2024 IEEE 33rd Conference on Electrical Performance of Electronic Packaging and Systems (EPEPS)*, IEEE, 2024, pp. 1–3.

[26] Song, S., and Sui, Y., "System Level Optimization for High-Speed SerDes: Background and the Road Towards Machine Learning Assisted Design Frameworks," *Electronics*, Vol. 8, No. 11, 2019. https://doi.org/10.3390/electronics8111233, URL https://www.mdpi.com/2079-9292/8/11/1233.

[27] Yang, Y.-T., and Hung, C.-M., "Heterogeneous Integration in Co-Packaged Optics," *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, 2025.